



WATER AND AIR TRANSPORT — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this chapter:

1. Kochi Port — Malabar Coast
2. Tides — Navigation, Fishing, Siltation Prevention
3. Cochin — India's Largest Shipyard
4. Tidal Ports Map — Kandla & Kolkata (Disputed)
5. Kandla — Gulf of Kutch Location
6. Chennai — Deep Port Myth (Actually Visakhapatnam)
7. Chennai — Artificial Port (Not Natural)
8. Kandla — Not on Eastern Coast
9. Paradip — Decongests Kolkata & Vishakhapatnam
10. Paradip — Located in Odisha
11. Paradip — India's Deepest Port
12. Marmugao — Located in Goa
13. Port-State Match (Haldia-WB, Paradip-Odisha, Kandla-Gujarat, Marmugao-Goa)
14. Kandla — Highest Import/Export Cargo Tonnage (2023-24)
15. Pondicherry UT Ports (Mahe, Karaikal, Pondicherry, Yanam)
16. Chennai — Not a Natural Harbour
17. Natural vs Artificial Ports (Vizag/Cochin Natural, Chennai/Tuticorin Artificial)
18. Kamarajar/Mundra/Vizag Port Facts
19. Eastern Coast Ports Sequence (Tuticorin-Chennai-Machilipatnam-Vizag)
20. Western Coast Ports Map (Veraval-Karwar-Tuticorin-Kakinada)
21. Kakinada — Andhra Pradesh Port City
22. Kolkata — India's Riverine Port (Hooghly River)
23. Kandla — No LNG Terminal (6 LNG Ports Elsewhere)
24. Kovalam — Three Crescent-Shaped Beaches
25. Sethusamudram Shipping Canal — 167km
26. Sethusamudram — Gulf of Mannar to Palk Bay
27. Dahej — Non-Major Port (12 Major Ports Listed)
28. Port-State Match (Kandla-Gujarat, Nhava Sheva-Maharashtra)
29. Port-State Match (Alleppey-Kerala, Ennore-TN, Kakinada-AP)
30. Krishnapatnam Port — Andhra Pradesh Beneficiary
31. Mumbai — India's Biggest Port
32. 13th Major Port — Vadnavan, Maharashtra
33. National Waterway 1 — Ganga-Bhagirathi-Hooghly (Allahabad-Haldia)
34. NW-1 Length — 1620 km, Longest in India (1986)
35. Kottapuram-Kollam — Kerala Coastal Canal Waterway
36. National Inland Navigation Institute — Patna
37. 12th Major Coal Port — Ennore, Near Chennai
38. Kandla — Gulf of Kutch Port (Post-Partition)
39. Jamnagar — Not a Gujarat Port Town
40. Bengaluru — Not a Harbour (Landlocked)
41. Chennai — India's Open Seaport
42. Haldia — 'Out Port' Example (Kolkata's Subsidiary)
43. Shipyard-State Match (Garden Reach-WB, Hindustan-AP, Mazgaon-Maharashtra)

- 44. Garden Reach Diesel Engine Plant — Ranchi
- 45. Cochin Airport — First Public Limited Company Airport
- 46. Raja Sansi International Airport — Amritsar
- 47. Karaikal — Eastern Coast (Not Western)
- 48. Alang, Gujarat — India's Largest Ship Breaking Yard
- 49. JNPT (Nhava Sheva) — Built to Decongest Mumbai Port (1989)
- 50. Jawaharlal Nehru Port — Located in Maharashtra
- 51. Rajasthan's Proposed Artificial Inland Port (Jalore)
- 52. Bengaluru — Not a Harbour (Repeat)
- 53. Mundra — Gujarat (Not Andhra Pradesh)
- 54. Panna — Non-Operational Airport in MP
- 55. MP International Airport Status — Disputed (All Domestic)
- 56. Kushinagar Airport — International Status (June 2020)
- 57. Air Transport in India — Started 1911 (Allahabad-Naini)
- 58. Cochin Airport — First Solar-Powered Airport in India
- 59. Cochin International Airport — World's First Fully Solar Airport

1. Kochi Port — Malabar Coast

- Kochi sea-port (Kerala) **Malabar Coast** par sthith graceful natural harbour hai — poore saal use hota hai. [Q1]

2. Tides — Navigation, Fishing, Siltation Prevention

- Tides ke sab 4 faayde sahi hain: **navigation-fishing** mein madadgar, high tide bade jahaz safely enter/exit karne dete hain, siltation rokhte hain, **Kandla-Diamond Harbour** tidal ports hain. [Q2]

3. Cochin — India's Largest Shipyard

- Cochin Shipyard Limited (CSL) **India** ki major shipbuilding-maintenance facility hai — India ka sabse bada shipyard. [Q3, Q4]

4. Tidal Ports Map — Kandla & Kolkata (Disputed)

- Map ports: **Kolkata-WB, Paradeep-Odisha, Cochin-Kerala, Kandla-Gujarat** — **Kandla aur Kolkata** dono tidal ports hain (disputed question hai, single answer possible nahi). [Q5]

5. Kandla — Gulf of Kutch Location

- Kandla (Deendayal) port **Kandla Creek** mein hai, **Gulf of Kutch** ke mouth se 90km door — protected natural harbour hai, India ka pehla SEZ hai. [Q6, Q9]

6. Chennai — Deep Port Myth (Actually Visakhapatnam)

- Cochin-Natural Port, JNPT-Mechanized Port, Kandla-Tidal Port sahi hain — '**Chennai-Deepest Port**' GALAT hai (asal mein Visakhapatnam sabse gehra landlocked port hai). [Q7]

⚠ EXAM TRAP — Visakhapatnam India ka sabse gehra port hai, Chennai NAHI

7. Chennai — Artificial Port (Not Natural)

- Chennai **kritrim (artificial) port** hai (1881 se commercial operations, India ka 3rd oldest major port) — Mormugao, Mumbai, Kochi natural ports hain. [Q8]

8. Kandla — Not on Eastern Coast

- Chennai, Paradip, Tuticorin eastern coast par hain — **Kandla western coast (Gujarat) par hai, eastern coast par NAHI.** [Q9]

9. Paradip — Decongests Kolkata & Vishakhapatnam

- Paradip port **Kolkata aur Vishakhapatnam ports ka traffic kam karne ke liye develop kiya gaya** — Odisha se Japan ko iron-ore export ke liye shuru hua tha. [Q10, Q11]

10. Paradip — Located in Odisha

- Paradip **Odisha mein sthith hai.** [Q12]

11. Paradip — India's Deepest Port

- Diye gaye options mein **Paradip India ka sabse gehra port hai (minimum depth se maximum vessel length decide hoti hai).** [Q13]

12. Marmugao — Located in Goa

- Marmugao seaport **Goa mein sthith hai.** [Q14]

13. Port-State Match (Haldia-WB, Paradip-Odisha, Kandla-Gujarat, Marmugao-Goa)

- Sahi match: **Haldia — West Bengal, Paradip — Odisha, Kandla (Deendayal) — Gujarat, Marmugao — Goa.** [Q15]

14. Kandla — Highest Import/Export Cargo Tonnage (2023-24)

- 2023-24 data ke mutabik **Kandla port sabse zyada import+export cargo tonnage handle karta hai (India ka main harbour).** [Q16]

15. Pondicherry UT Ports (Mahe, Karaikal, Pondicherry, Yanam)

- Puducherry UT ke 4 sub-areas ke ports: **Mahe, Karaikal, Pondicherry, Yanam** — 1954 tak French rule tha. [Q17]

16. Chennai — Not a Natural Harbour

- Chennai **India ka sabse purana aur sabse bada artificial harbour hai** — petroleum products, fertilizers, iron-ore, general cargo handle karta hai. [Q18]

17. Natural vs Artificial Ports (Vizag/Cochin Natural, Chennai/Tuticorin Artificial)

- Vishakhapatnam aur Cochin **natural harbours hain** — Chennai aur Tuticorin artificial ports hain. [Q19]

18. Kamarajar/Mundra/Vizag Port Facts

- Sahi facts: **Kamarajar Port (ex-Ennore) India ka pehla company-registered major port hai, Mundra Port India ka sabse bada privately-owned port hai (Gujarat) — 'Vishakhapatnam largest container port hai' GALAT hai (Jawaharlal Nehru Port/Nhava Sheva India ka largest container port hai).** [Q20]

⚠ EXAM TRAP — JNPT (Nhava Sheva) India ka largest container port hai, Vishakhapatnam NAHI

19. Eastern Coast Ports Sequence (Tuticorin-Chennai-Machilipatnam-Vizag)

- Eastern coast ports sequence: **Tuticorin (TN), Chennai (TN), Machilipatnam (AP), Vishakhapatnam (AP).** [Q21]

20. Western Coast Ports Map (Veraval-Karwar-Tuticorin-Kakinada)

- Map match: **A-Veraval, B-Karwar, C-Tuticorin, D-Kakinada** — Mangalore port Karnataka mein hai. [Q22]

21. Kakinada — Andhra Pradesh Port City

- Kakinada Andhra Pradesh ka harbour city hai — Anchorage Port, Deep Water Port, Fishing Harbour, Ship-Breaking Unit shamil hain. [Q23]

22. Kolkata — India's Riverine Port (Hooghly River)

- Map ke 4 ports: **Kandla, Kochi, Paradip, Kolkata** — Kolkata port riverine hai (Hooghly river ke kinare). [Q24]

23. Kandla — No LNG Terminal (6 LNG Ports Elsewhere)

- India ke 6 LNG terminals: **Dahej, Hazira, Cochin, Dabhol, Ennore, Mundra** — Kandla mein LNG terminal NAHI hai. [Q25]

24. Kovalam — Three Crescent-Shaped Beaches

- 3 crescent-shaped beaches **Kovalam (Kerala)** mein milte hain. [Q26]

25. Sethusamudram Shipping Canal — 167km

- Sethusamudram Shipping Canal Project (India-Sri Lanka ke beech shallow straits mein) **167 km lambi hai** — Palk Strait (Bay of Bengal) ko Gulf of Mannar se jodta hai. [Q27]

26. Sethusamudram — Gulf of Mannar to Palk Bay

- Sethusamudram project **Gulf of Mannar ko Palk Bay se jodta hai** — Indian peninsula ke around continuous navigable route provide karta hai. [Q28]

27. Dahej — Non-Major Port (12 Major Ports Listed)

- Dahej (Gulf of Khambhat, Bharuch, Gujarat) **non-major port hai** — India ke **12 major ports: Kolkata, Paradip, Visakhapatnam, Chennai, Tuticorin, Kochi, New Mangalore, Marmugoa, Mumbai, JNPT (Nhava Sheva), Ennore (Kamarajar), Kandla (Deendayal).** [Q29]

28. Port-State Match (Kandla-Gujarat, Nhava Sheva-Maharashtra)

- Match: Kandla (Deendayal) — Gujarat, Nhava Sheva — Maharashtra, Paradip — Odisha, Tuticorin (V.O. Chidambarnar) — Tamil Nadu. [Q30]

29. Port-State Match (Alleppey-Kerala, Ennore-TN, Kakinada-AP)

- Match: Alleppey — Kerala, Ennore (Kamarajar) — Tamil Nadu, Paradip — Odisha, Kakinada — Andhra Pradesh. [Q31]

30. Krishnapatnam Port — Andhra Pradesh Beneficiary

- Krishnapatnam port se Andhra Pradesh ko sabse zyada faayda hoga (ye port AP mein hai). [Q32]

31. Mumbai — India's Biggest Port

- Mumbai Port natural deep-water harbour hai, India ka sabse bada port (west coast par) — industrial development mein bada role tha. [Q33, Q34]

32. 13th Major Port — Vadhavan, Maharashtra

- India ka 13th major port Vadhavan (Maharashtra) mein banega — 'landlord model' par develop hoga. [Q35]

33. National Waterway 1 — Ganga-Bhagirathi-Hooghly (Allahabad-Haldia)

- Ganga River ka Allahabad (Prayagraj) se Haldia tak ka hissa National Waterway-1 ghoshit hai. [Q36, Q37, Q38]

34. NW-1 Length — 1620 km, Longest in India (1986)

- NW-1 1986 mein establish hui — total length 1620 km, India ka sabse lamba internal waterway — 'Haldia to Allahabad' hi sahi answer hai. [Q39]

35. Kottapuram-Kollam — Kerala Coastal Canal Waterway

- Kottapuram se Kollam tak Kerala Coastal Canal (West Coast Canal) National Waterway hai — 1992 Act se declare hui. [Q40, Q41]

36. National Inland Navigation Institute — Patna

- NINI (National Inland Navigation Institute) Patna, Bihar mein IWAI (Inland Waterways Authority of India) ne setup ki. [Q42]

37. 12th Major Coal Port — Ennore, Near Chennai

- India ka 12th major coal-handling port Ennore (Chennai ke paas) mein bana — country ka pehla incorporated port hai. [Q43]

38. Kandla — Gulf of Kutch Port (Post-Partition)

- Kandla (Deendayal) Gulf of Kutch par sthith hai — 1950s mein Karachi port Pakistan mein chale jaane ke baad Western India ke main seaport ke roop mein bana. [Q44]

39. Jamnagar — Not a Gujarat Port Town

- Okha, Porbandar, Alang, Bedi, Bhavnagar, Dahej, Jafarabad, Mandvi, Navlayi, Surat, Veraval **Gujarat** ke port towns hain — Jamnagar port town **NAHI** hai (Gulf of Kutch ke south mein sthith hai). [Q45]

⚠ EXAM TRAP — Jamnagar — Gujarat ka port town NAHI hai (Kutch ki khaadi ke south mein sthith hai)

40. Bengaluru — Not a Harbour (Landlocked)

- Kandla, Cochin, Mangalore eminent harbours hain — Bengaluru landlocked city hai, koi port **NAHI**. [Q46]

41. Chennai — India's Open Seaport

- Open seaport (jahan jahaz khule paani mein coast karte hain, quays coast ke parallel bante hain) **India** mein **Chennai** hai. [Q47]

42. Haldia — 'Out Port' Example (Kolkata's Subsidiary)

- Haldia port (Kolkata se 90km door) 'Out Port' ka specific example hai — deep water mein establish hota hai, main port ka subsidiary port hai. [Q48]

43. Shipyard-State Match (Garden Reach-WB, Hindustan-AP, Mazgaon-Maharashtra)

- Match: Garden Reach — West Bengal, Hindustan Shipyard — Andhra Pradesh, Mazgaon Docks — Maharashtra, Cochin Shipyard — Kerala. [Q49]

44. Garden Reach Diesel Engine Plant — Ranchi

- Garden Reach Shipbuilders and Engineering Limited ka diesel engine plant **Ranchi** mein hai. [Q50]

45. Cochin Airport — First Public Limited Company Airport

- Cochin Airport **India** ka pehla public-limited-company-owned airport hai — May 1999 mein K.R. Narayanan ne inaugurate kiya, June 1999 se commercial service — Nedumbassery mein hai, Kerala ka sabse busy/bada airport. [Q51]

46. Raja Sansi International Airport — Amritsar

- Raja Sansi International Airport (Sri Guru Ram Das Jee International Airport) **Amritsar** mein hai — Guru Ram Das ke naam par. [Q52]

47. Karaikal — Eastern Coast (Not Western)

- Daman, Janjira, Ratnagiri western coast par hain — **Karaikal** eastern coast par hai (western coast par **NAHI**). [Q53]

48. Alang, Gujarat — India's Largest Ship Breaking Yard

- Alang (Bhavnagar district, Gujarat) **India ka sabse bada ship-breaking yard hai** — world ke ~50% ship-salvage yahan recycle hota hai. [Q54]

49. JNPT (Nhava Sheva) — Built to Decongest Mumbai Port (1989)

- Jawaharlal Nehru Port (Nhava Sheva) **1989 mein Mumbai Port ka pressure kam karne ke liye banaya gaya**. [Q55]

50. Jawaharlal Nehru Port — Located in Maharashtra

- JNP (Nhava Sheva Port) **Maharashtra mein sthith hai**. [Q56]

51. Rajasthan's Proposed Artificial Inland Port (Jalore)

- Rajasthan Government **Jalore mein artificial inland port develop karne ki soch rahi hai** — Gujarat se **Arabian Sea ka paani laake** — navigation channel 300m chaudi, 25m gehri hogi, Kori Creek (Lakpat) se Rann of Kutch se hoke Jalore tak. [Q57]

52. Bengaluru — Not a Harbour (Repeat)

- Cochin, Mangalore, Kandla harbours hain — **Bengaluru harbour NAHI hai (landlocked)** — India mein 12 major aur ~200 non-major/intermediate ports hain. [Q58]

53. Mundra — Gujarat (Not Andhra Pradesh)

- Marmugao-Goa, Paradip-Odisha, Mangalore-Karnataka sahi hain — **'Mundra-Andhra Pradesh' GALAT hai (Mundra Gujarat mein hai)**. [Q59]

EXAM TRAP — Mundra — Gujarat mein hai, Andhra Pradesh mein NAHI

54. Panna — Non-Operational Airport in MP

- Khajuraho, Jabalpur, Gwalior operational airports hain — **Panna ka airport non-operational hai**. [Q60]

55. MP International Airport Status — Disputed (All Domestic)

- Bhopal, Indore, Khajuraho, Gwalior — **sab domestic airports hain (koi international NAHI)**, isliye MPPSC ne is question ko delete kar diya. [Q61]

56. Kushinagar Airport — International Status (June 2020)

- 24 June 2020 ko Union Cabinet ne **Kushinagar Airport (UP) ko international status diya**. [Q62]

57. Air Transport in India — Started 1911 (Allahabad-Naini)

- India mein air transport ki shuruaat **1911 mein Allahabad (Prayagraj) aur Naini ke beech flight se maani jaati hai**. [Q63]

58. Cochin Airport — First Solar-Powered Airport in India

- Cochin International Airport Ltd (CIAL) **duniya ka pehla poori tarah solar-power se chalne wala airport hai (2015 mein achieve kiya).** *[Q64]*

| 59. Cochin International Airport — World's First Fully Solar Airport

- Cochin International Airport **PPP model ke tahat develop hua India ka pehla airport hai — dedicated solar plant ke saath world ka pehla fully-solar-powered airport bana.** *[Q65]*